```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
np.random.seed(42)
normal_data = np.random.normal(loc=50, scale=5, size=1000)
skewed_data = np.random.exponential(scale=5, size=1000)
df = pd.DataFrame({
    "Normal_Data": normal_data,
    "Skewed_Data": skewed_data
})
plt.figure()
sns.histplot(df["Normal_Data"], kde=True)
plt.title("Normal Distribution (Mean=50, SD=5)")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
plt.figure()
sns.histplot(df["Skewed_Data"], kde=True)
plt.title("Exponential Distribution (Right Skewed)")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
normal_skew = stats.skew(normal_data)
skewed_skew = stats.skew(skewed_data)
print("Skewness Values:")
```

```python
print("Normal Data:", normal_skew)
print("Skewed Data:", skewed_skew)
normal_kurt = stats.kurtosis(normal_data)
skewed_kurt = stats.kurtosis(skewed_data)
print("\nKurtosis Values (Fisher's Excess Kurtosis):")
print("Normal Data:", normal_kurt)
print("Skewed Data:", skewed_kurt)
def calculate_z_scores(data):  1 usage
    mean = np.mean(data)
    std = np.std(data)
    return (data - mean) / std
z_scores = calculate_z_scores(skewed_data)
outliers = skewed_data[np.abs(z_scores) > 3]
print("\nNumber of Outliers in Skewed Data (|Z| > 3):", len(outliers))
print("Outlier Values:\n", outliers)
```

```
Skewness Values:
Normal Data: 0.11680083110533707
Skewed Data: 1.9808119458203122

Kurtosis Values (Fisher's Excess Kurtosis):
Normal Data: 0.06620589292148482
Skewed Data: 5.37944048539541

Number of Outliers in Skewed Data (|Z| > 3): 20
Outlier Values:
 [28.04315616 21.97031946 21.57375542 37.20861455 22.02566582 23.69693404
 25.69748337 24.45327481 24.24131633 26.35965988 20.57649039 28.56439943
 20.88963298 24.0241772  28.04373115 32.0273644  20.99507862 20.52848561
 20.71042688 26.94944054]
```

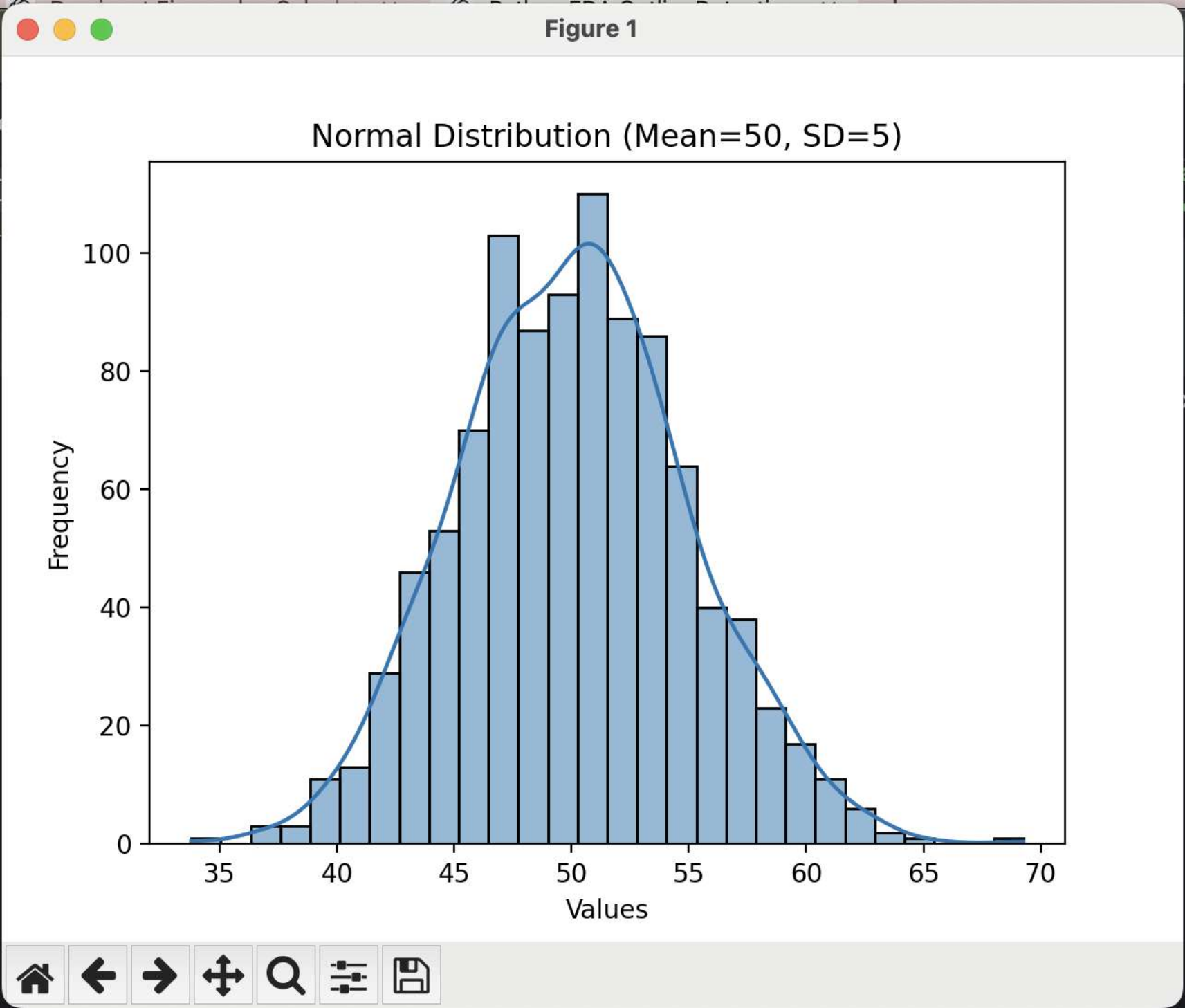
Figure 1
Normal Distribution (Mean=50, SD=5)
Frequency
Values
0
20
40
60
80
100
35
40
45
50
55
60
65
70

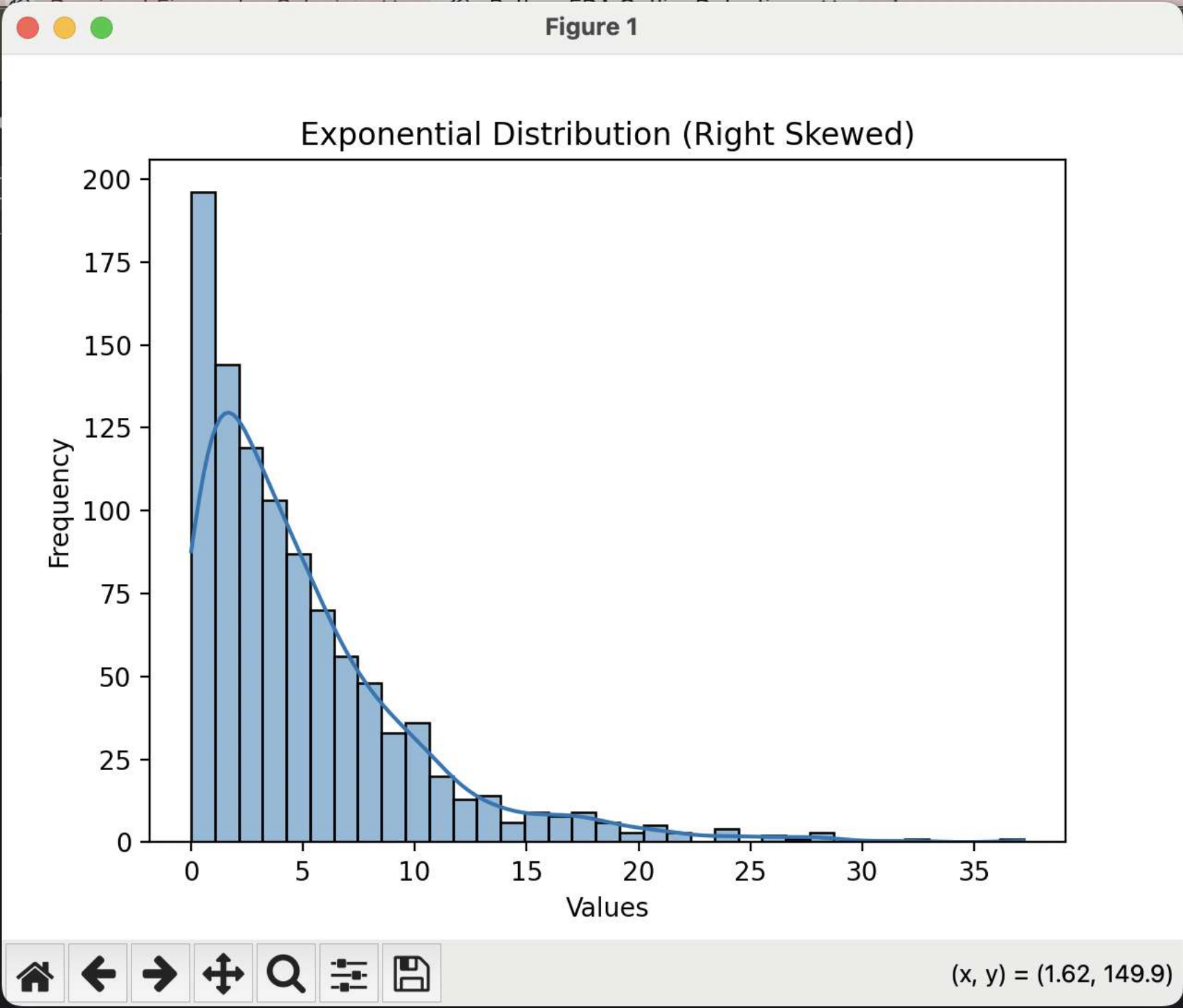
Figure 1
Exponential Distribution (Right Skewed)
Frequency
Values
0
25
50
75
100
125
150
175
200
0
5
10
15
20
25
30
35
(x, y) = (1.62, 149.9)